

Computational Sciences

Basic Git Commands II

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1 Check changes to files

1. Modify README.md and check the status:

```
git status
```

2. View changes:

```
git diff
```

3. Stage and commit:

```
git add README.md
```

```
git commit -m "Updated README with new line"
```

2 Exploring the History

1. View commit history:

```
git log
```

2. Use concise format:

```
git log --oneline
```

3 Undoing Changes

1. Assume you have edited or deleted the file `README.md`, but you have not yet used `git add` to add the changes to the staging area. You want to undo these local changes and return the file to its original state, then use

```
git checkout -- README.md
```

2. Unstage changes (cancel the `git add` command):

```
git reset HEAD README.md
```

4 Branching

1. Create and switch to a new branch:

```
git branch secondary  
git checkout secondary
```

2. Create a new file `new-file.txt`.
3. Add, commit and push a file on a branch:

```
git add new-file.txt  
git commit -m "Added new file on secondary branch"
```

```
git push origin secondary
```

5 Merge branches

This exercise will guide you through the process of using `git merge` to integrate changes from one branch into another. Imagine you are working on a project with the `main` branch.

1. Be sure to be in the `main` branch with the command

```
git checkout main
```

2. Create a text file `file.txt`.
3. Add, commit and push on `main`:

```
git add file.txt
git commit -m "commit on main"
git push origin main
```

4. Create a new branch from `main` and switch to it

```
git branch secondary
git checkout secondary
```

5. Edit the file `file.txt` in the secondary branch, then add, commit and push on the secondary branch:

```
git add file.txt
git commit -m "commit on secondary"
git push origin secondary
```

6. Switch to the `main` branch. You will notice that it contains the original version of the file:

```
git checkout main
```

7. To merge the secondary branch with the main branch, run:

```
git merge secondary
```

Now the file `file.txt` in the main branch is equal to the file in the secondary branch. Add, commit and push on main to update also the remote repository.

```
git add file.txt
git commit -m "commit on main"
git push origin main
```

6 Practice with `.gitignore`

1. Create a `.gitignore` file and add directories or files that you have in the local directory, but you do not want to upload on the remote repository.